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5CC507 Databases

Database Design II

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Aims of the Session

This session aims to help you:

1. Understand the basic principles underpinning the design process of a relational database

2. Learn the basic aspects of the E/R model

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History of ER models

ER diagrams are visual tools that are helpful to represent the ER model. Peter Chen proposed ER Diagram in 1971 to create a uniform convention that can be used for relational databases and networks. He aimed to use an ER model as a conceptual modeling approach

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Why use ER Diagrams?

Here, are prime reasons for using the ER Diagram

- Helps you to define terms related to entity relationship modeling
- Provide a preview of how all your tables should connect, what fields are going to be on each table
- Helps to describe entities, attributes, relationships
- ER diagrams are translatable into relational tables which allows you to build databases quickly
- ER diagrams can be used by database designers as a blueprint for implementing data in specific software applications
- The database designer gains a better understanding of the information to be contained in the database with the help of ERP diagram
- ERD Diagram allows you to communicate with the logical structure of the database to users

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ER Diagrams Symbols & Notations

- **Rectangles:** This Entity Relationship Diagram symbol represents entity types
- **Ellipses :** Symbol represent attributes
- **Diamonds:** This symbol represents relationship types
- **Lines:** It links attributes to entity types and entity types with other relationship types
- **Primary key:** attributes are underlined
- **Double Ellipses:** Represent multi-valued attributes

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ER Diagrams Symbols & Notations

Components of the ER Diagram

This model is based on three basic concepts:

✓ Entities
✓ Attributes
✓ Relationships

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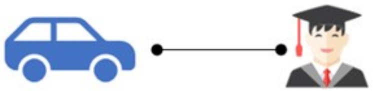
Design Process

Entity

Person,place,object,event or concept about which data is to be maintained
Example: Car, Student

Attribute

Property or characteristic of an entity
Example: Color of car Entity
Name of Student Entity



Relation

Association between the instances of one or more entity types
Example: Blue Car Belongs to Student Jack

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WHAT IS ENTITY?

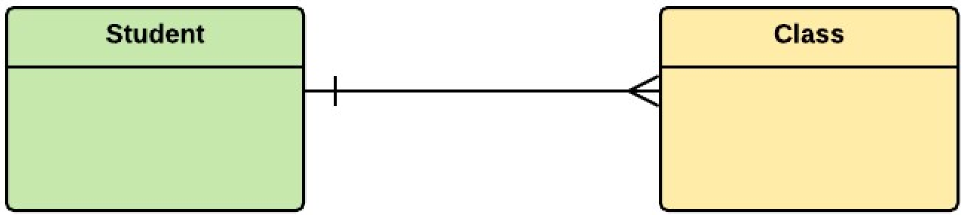
A real-world thing either living or non-living that is easily recognizable and nonrecognizable. It is anything in the enterprise that is to be represented in our database. It may be a physical thing or simply a fact about the enterprise or an event that happens in the real world.

An entity can be place, person, object, event or a concept, which stores data in the database. The characteristics of entities are must have an attribute, and a unique key. Every entity is made up of some 'attributes' which represent that entity.

WHAT IS ENTITY?

Examples of entities:

- **Person:** Employee, Student, Patient
- **Place:** Store, Building
- **Object:** Machine, product, and Car
- **Event:** Sale, Registration, Renewal
- **Concept:** Account, Course



Relationship

Relationship is nothing but an association among two or more entities.

E.g., Tom works in the Chemistry department

Entities take part in relationships. We can often identify relationships with verbs or verb phrases.

For example:

- You are attending this lecture
- I am giving the lecture
- Just loke entities, we can classify relationships according to relationship-types:

For example:

- A student attends a lecture
- A lecturer is giving a lecture.

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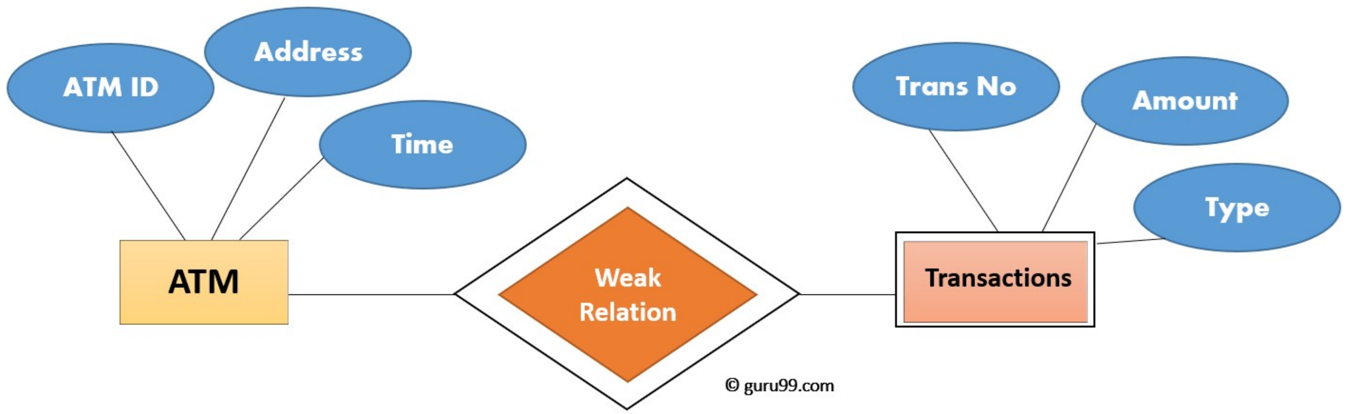
Weak Entities

A weak entity is a type of entity which doesn't have its key attribute. It can be identified uniquely by considering the primary key of another entity. For that, weak entity sets need to have participation.



Strong vs Weak Entities

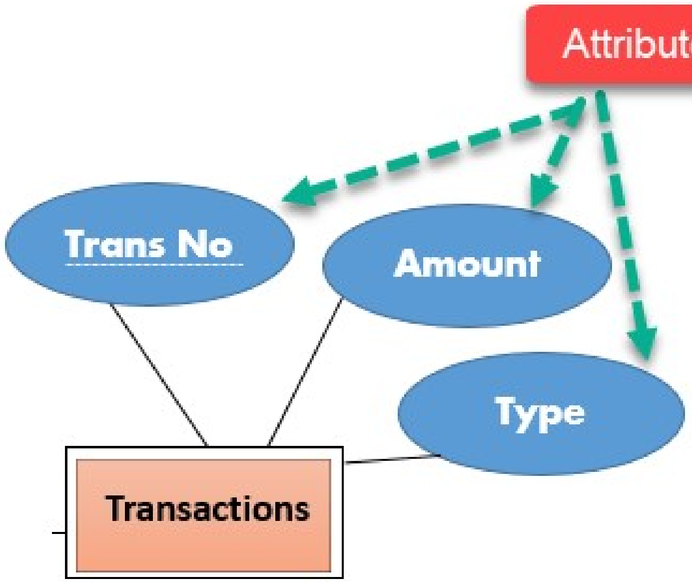
Strong Entity Set	Weak Entity Set
Strong entity set always has a primary key.	It does not have enough attributes to build a primary key.
It is represented by a rectangle symbol.	It is represented by a double rectangle symbol.
It contains a Primary key represented by the underline symbol.	It contains a Partial Key which is represented by a dashed underline symbol.
The member of a strong entity set is called as dominant entity set.	The member of a weak entity set called as a subordinate entity set.
Primary Key is one of its attributes which helps to identify its member.	In a weak entity set, it is a combination of primary key and partial key of the strong entity set.
In the ER diagram the relationship between two strong entity set shown by using a diamond symbol.	The relationship between one strong and a weak entity set shown by using the double diamond symbol.
The connecting line of the strong entity set with the relationship is single.	The line connecting the weak entity set for identifying relationship is double.

Attributes

It is a single-valued property of either an entity-type or a relationship-type.

For example, a lecture might have attributes: time, date, duration, place, etc.

An attribute in ER Diagram examples, is represented by an Ellipse



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Attributes

Types of Attributes	Description
Simple attribute	Simple attributes can't be divided any further. For example, a student's contact number. It is also called an atomic value.
Composite attribute	It is possible to break down composite attribute. For example, a student's full name may be further divided into first name, second name, and last name.
Derived attribute	This type of attribute does not include in the physical database. However, their values are derived from other attributes present in the database. For example, age should not be stored directly. Instead, it should be derived from the DOB of that employee.
Multivalued attribute	Multivalued attributes can have more than one values. For example, a student can have more than one mobile number, email address, etc.

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Cardinality

Defines the numerical attributes of the relationship between two entities or entity sets.

- Different types of cardinal relationships are:
  - One-to-One Relationships
  - One-to-Many Relationships
  - May to One Relationships
  - Many-to-Many Relationships

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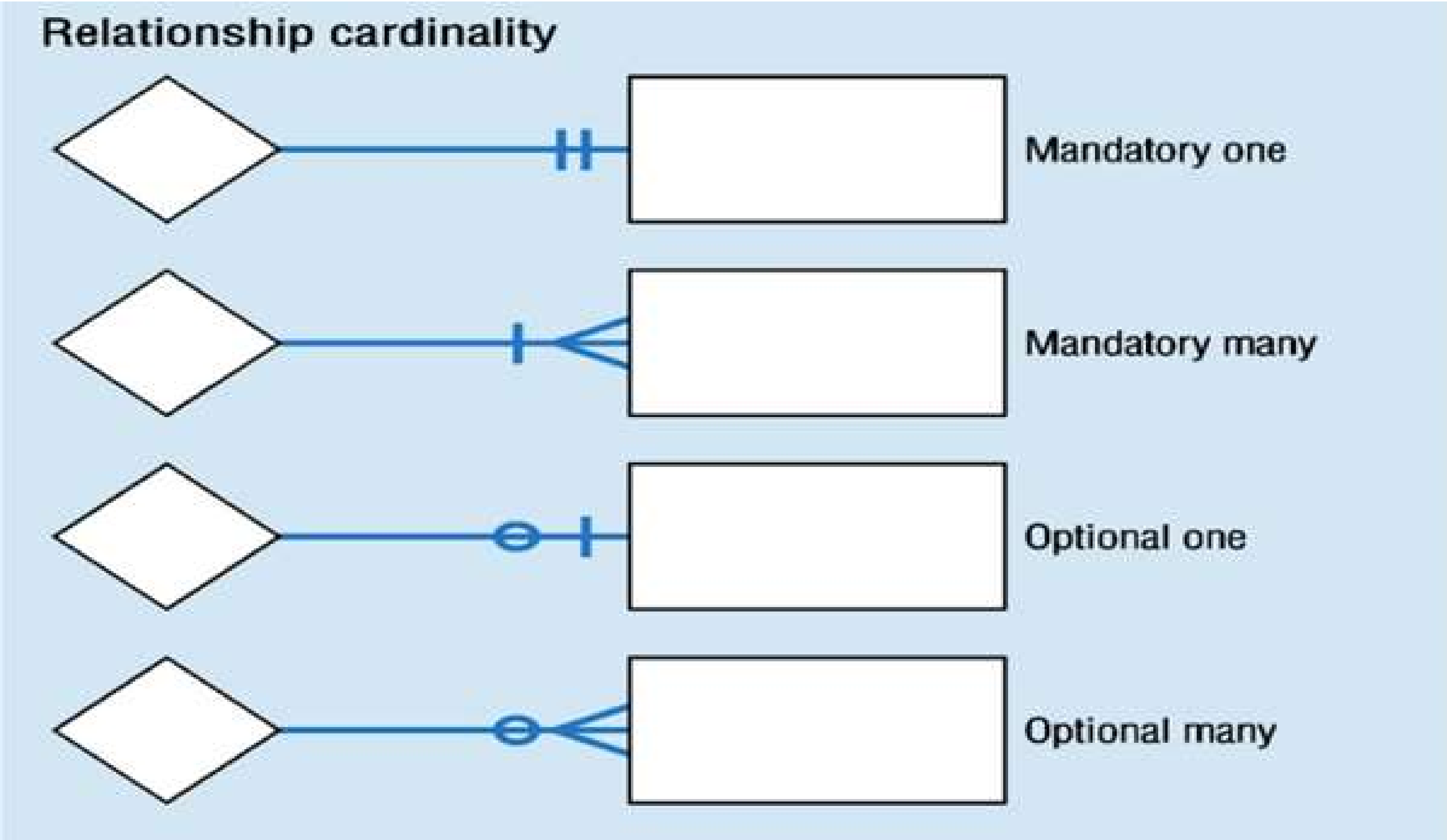
Cardinality



Entity Relationship Diagram Example:

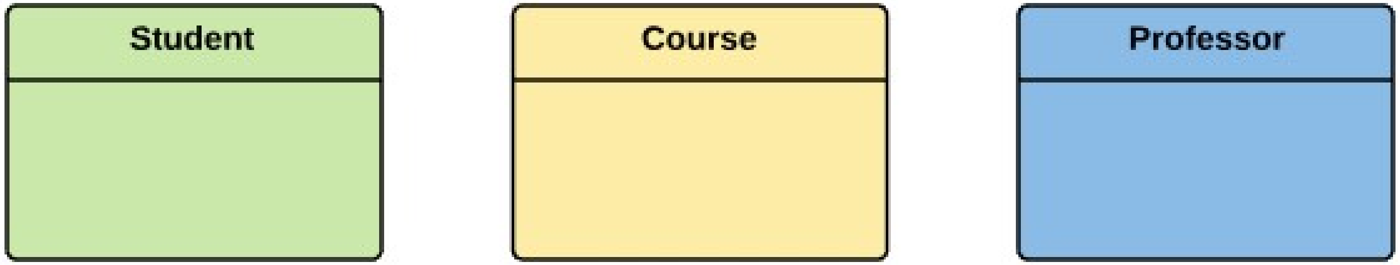
In a university, a Student enrolls in Courses. A student must be assigned to at least one or more Courses. Each course is taught by a single Professor. To maintain instruction quality, a Professor can deliver only one course

How to Create an Entity Relationship Diagram (ERD)

Step 1 Entity Identification

We have three entities

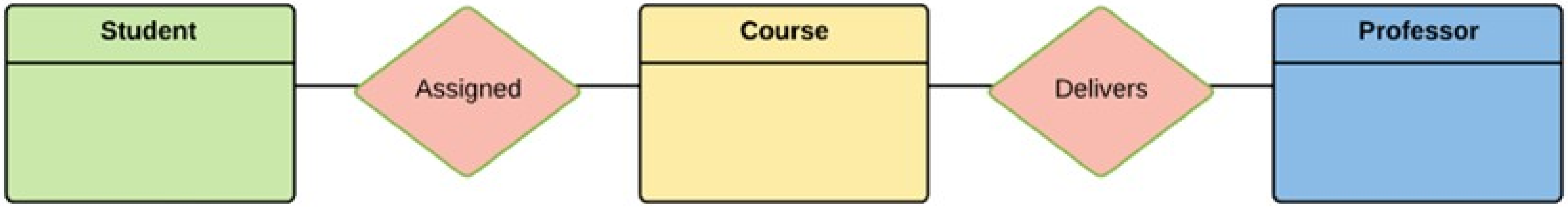
- Student
- Course
- Professor



Step 2 Relationship Identification

We have the following two relationships

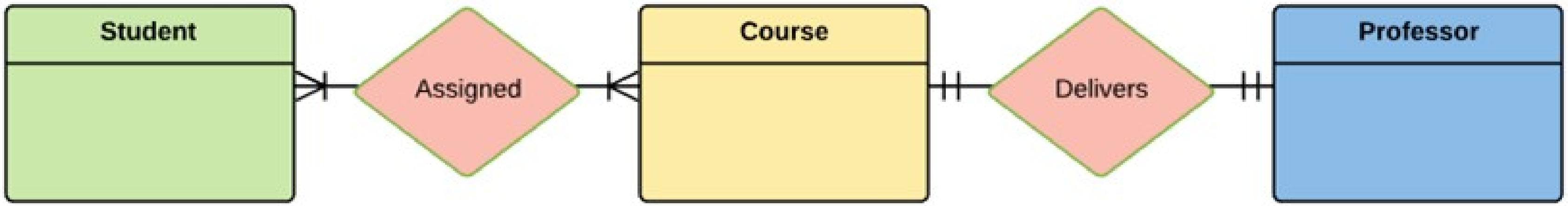
- The student is assigned a course
- Professor delivers a course



Step 3 Cardinality Identification

For them problem statement we know that,

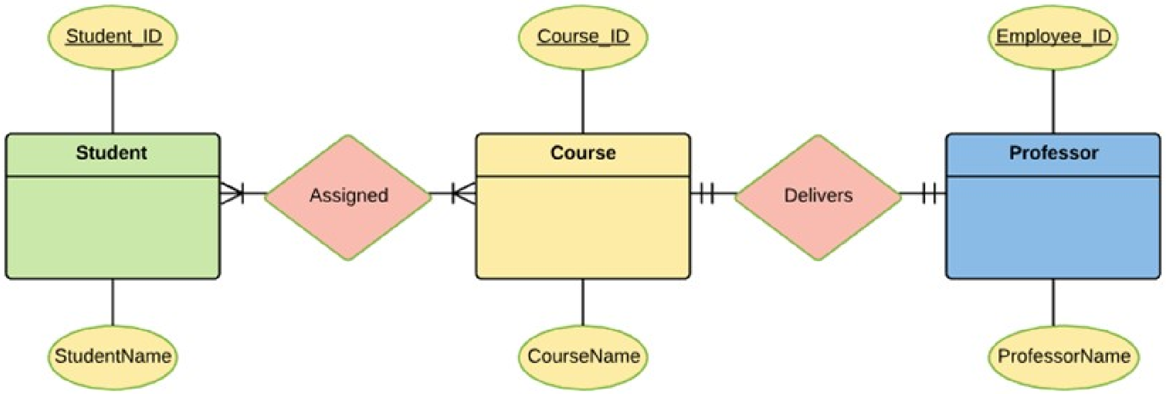
- A student can be assigned multiple courses
- A Professor can deliver only one course



Step 4 Identify Attributes

You need to study the files, forms, reports, data currently maintained by the organization to identify attributes

Once, you have a list of Attributes, you need to map them to the identified entities. Ensure an attribute is to be paired with exactly one entity. If you think an attribute should belong to more than one entity, use a modifier to make it unique.



Step 5 Create the ERD Diagram

A more modern representation of Entity Relationship Diagram

Best Practices for Developing Effective ER Diagrams

Here are some best practice or example for Developing Effective ER Diagrams.

- Eliminate any redundant entities or relationships
- You need to make sure that all your entities and relationships are properly labeled
- There may be various valid approaches to an ER diagram. You need to make sure that the ER diagram supports all the data you need to store
- You should assure that each entity only appears a single time in the ER diagram
- Name every relationship, entity, and attribute are represented on your diagram
- Never connect relationships to each other
- You should use colors to highlight important portions of the ER diagram

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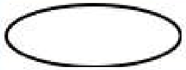
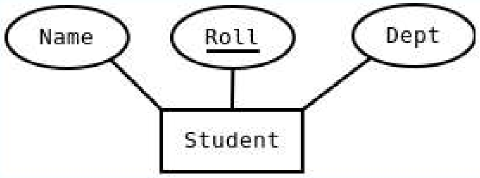
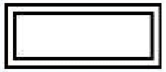
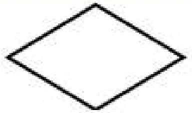
Summary

Term	Notation	Remarks
Entity set		Name of the set is written inside the rectangle
Attribute		Name of the attribute is written inside the ellipse
Entity with attributes		Roll is the primary key; denoted with an underline
Weak entity set		
Relationship set		Name of the relationship is written inside the diamond
Related enity sets		
Relationship cardinality		A person can own zero or more cars but no two persons can own the same car
Relationship with weak entity set